University of Stavanger

EXAM IN: STA510 STATISTICAL MODELING AND SIMULATION

DURATION: 4 HOURS DATE: FEBRUARY 12, 2024
PERMITTED AIDS: Approved simple calculator
THE EXAM CONSISTS OF 5 PROBLEMS ON 7 PAGES, 9 PAGES OF ENCLOSURES.
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Problem 1

Consider the function

$$f(x) = k(x-2)^2, \quad 0 \leq x \leq 2.$$

a) Show that $f(x)$ is a probability density function if $k = \frac{3}{8}$. (4P)

b) Assume we have a random variable X with probability density function $f(x)$.
Find $P(X \leq 1)$. (3P)
Find the expectation of X. (3P)
Let $Y = \frac{X_1+X_2-2}{4}$ where X_1 and X_2 are random variables with probability density function $f(x)$. Find the expectation of Y. (3P)

Problem 2

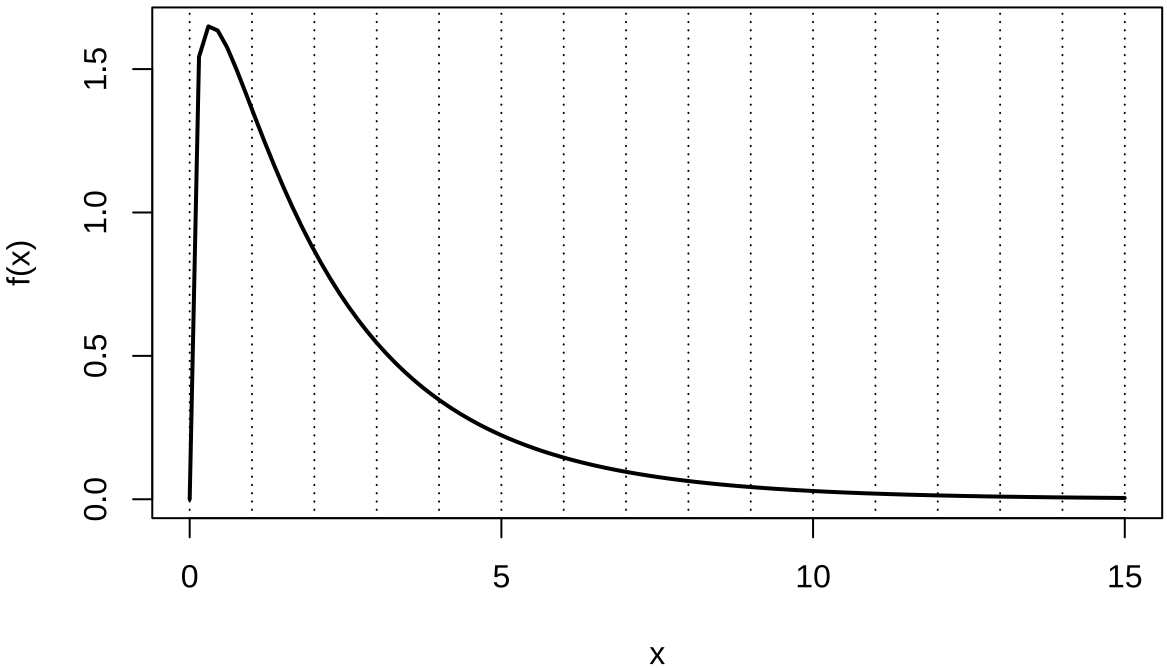
Consider a random variable X with probability density

$$f(x) = 6x(1-x),\ 0 \leq x \leq 1.$$

a) Write down the algorithm (with words and formulas, not code) to simulate a random variable with density $f(x)$ with the acceptance-rejection method. Specify also potential constants of the algorithm. (8P)

Problem 3

Consider the function $g(x) = \frac{1}{2}e^{-x^{3/4}+2}x^{1/3}$ which is plotted below



Now, we want to estimate the integral of

$$\theta = \int_1^6 \frac{1}{2}e^{-x^{3/4}+2}x^{1/3}dx$$

Consider the following R-code:

```
gx <- function(x)
  exp(-x^(3/4)+2) * x^(1/3) /2

Nsim <- 10000

# Estimation 1
x1 <- runif(Nsim/2,1,6)
x2 <- 5-x1
Est1 <- 5*mean(gx(c(x1,x2)))

# Estimation 2
x <- rgamma(Nsim,4,1.4)
Est2 <- mean(dgamma(x,4,1.4)/(gx(x)*(x<6 & x>1)))

# Estimation 3
x <- rgamma(Nsim,4,1.4)
Est3 <- mean(gx(x)*(x<6 & x>1)/dgamma(x,4,1.4))
```

a) Which of the above estimates correctly estimate the integral θ? Argue for each estimate which algorithm has been implemented and why it produces correct or wrong results. (4+4+3P)

b) Does importance sampling always lead to a reduction of the variance of the estimator? Justify your answer. (3P)

Problem 4

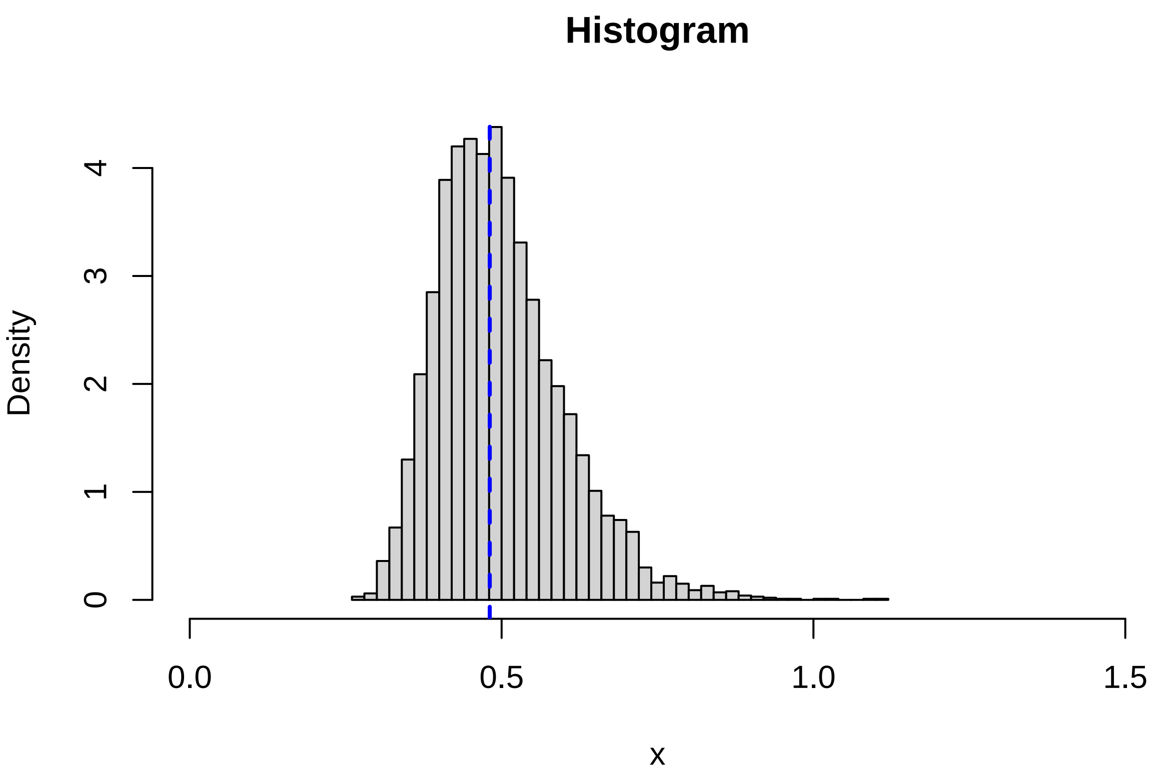
Suppose the lifetimes of a system follow a probability distribution with parameter θ. Assume we observe the following lifetimes $t_i, i = 1, \ldots, 20$ (in years)

$$0.62,\ 0.61,\ 1.64,\ 0.42,\ 0.45,\ 0.98,\ 0.09,\ 0.98,\ 1.91,\ 0.13$$
$$0.46,\ 0.39,\ 1.10,\ 0.55,\ 0.28,\ 0.35,\ 0.83,\ 0.37,\ 0.96,\ 1.03$$

and an estimator for θ is given by

$$\hat{\theta} = \frac{n}{\sum_{i=1}^{n}(t_i + 2t_i^{3/2})} = 0.481.$$

By bootstrap we create the following plot



a) Describe the algorithm to estimate the distribution of $\hat{\theta}$. (4P)

Below is a set of summary statistics of the bootstrap.

```
print(round(mean(thetaest1B),3))
```

```
## [1] 0.498
```

```
print(round(median(thetaest1B),3))
```

```
## [1] 0.485
```

```
print(round(sd(thetaest1B),3))
```

```
## [1] 0.101
```

```
print(round(quantile(thetaest1B,
                probs=c(0.01,0.025,0.05,0.1,0.9,0.95,0.975,0.99)),3))
```

```
##    1%  2.5%    5%   10%   90%   95% 97.5%   99%
## 0.321 0.343 0.361 0.383 0.631 0.686 0.727 0.801
```

b) Find the variance and the MSE of the estimator. (4P)
Find a 90% standard normal bootstrap interval. (4P)
Find a 90% percentile bootstrap interval. Comment on any differences (or lack thereof) between the intervals. (2+2P)

Problem 5

The occurrence of earthquakes can be modeled as a homogeneous Poisson process with intensity parameter λ where λ is the expected number of earthquake per year. Let us assume we live in a region with $\lambda = 10$. In a homogenous Poisson process, the time between earthquakes $T_1, \dots, T_n$ follows an exponential distribution. Let us assume $T_1, \dots, T_n \sim Exp(a \cdot b)$, i.e. the rate parameter $\lambda = a \cdot b$ and $E(T_i) = \frac{1}{a \cdot b}$. Further, we have recorded $n = 20$ observations $t_i, i = 1, \dots, 20$

$$0.08,\ 0.12,\ 0.01,\ 0.01,\ 0.04,\ 0.29,\ 0.12,\ 0.05,\ 0.10,\ 0.01$$
$$0.14,\ 0.08,\ 0.12,\ 0.44,\ 0.11,\ 0.10,\ 0.19,\ 0.07,\ 0.03,\ 0.06$$

and we assume the prior is given by $a \sim Exp(3)$ and $b \sim Exp(3)$.

a) Show that the posterior can be written as $p(a, b; t_1, \dots, t_n) \propto (a \cdot b)^n e^{-ab \sum_{i=1}^n t_i} e^{-3(a+b)}$. (4P)

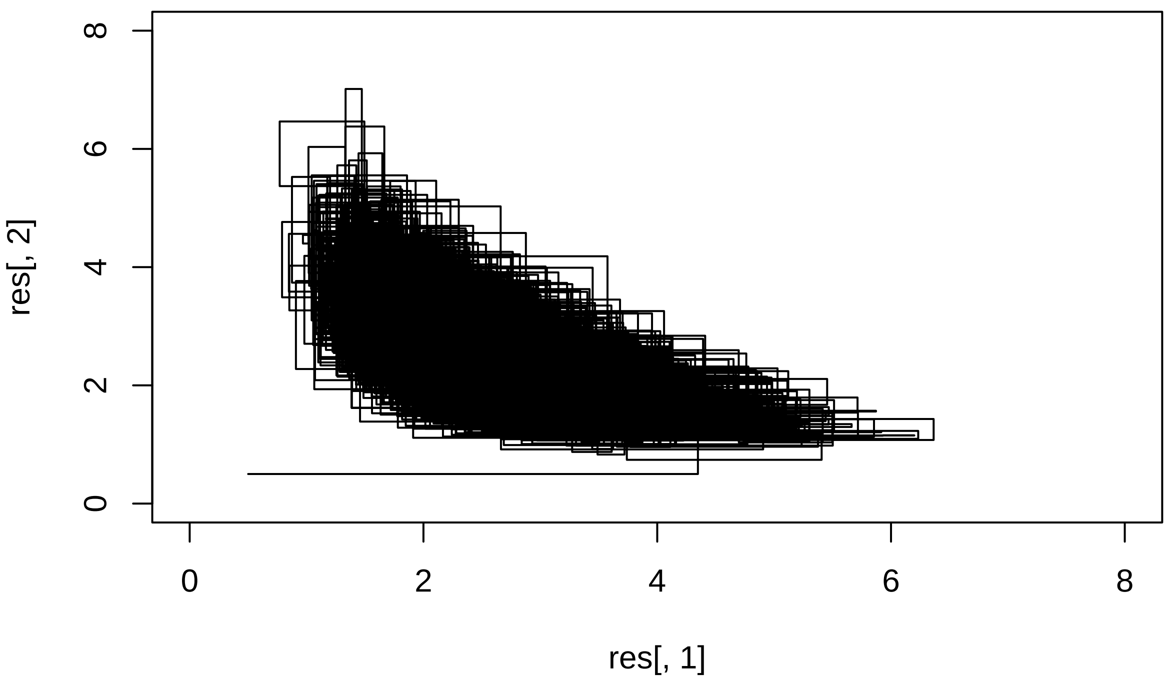
b) Find the distributions for the conditional posteriors $a|b, t_1, \dots, t_n$ and $b|a, t_1, \dots, t_n$.
Hint: If $\log p(x) = a + b \log(x) + cx$ for $x > 0$ then $x \sim Gamma(b + 1, -1/c)$. (5+2P)

Now we want to sample from the posterior distribution $p(a, b; t_1, \ldots, t_n)$. The first algorithm, and a plot of output are given by

```
data <- c(0.08, 0.12, 0.01, 0.01, 0.04, 0.29, 0.12, 0.05, 0.10, 0.01,
          0.14, 0.08, 0.12, 0.44, 0.11, 0.10, 0.19, 0.07, 0.03, 0.06)

sdata <- sum(data)
n <- length(data)

Nsim <- 10000
lambda <- c(0.5,0.5)
res <- matrix(NA, nrow = 2*Nsim+1, ncol = 2)
res[1,] <- lambda
k <- 2
set.seed(1)
for (i in 1:Nsim){
  lambda[1] <- rgamma(1, shape = n+1, scale = 1/(lambda[2]*sdata+3))
  res[k,] <- lambda
  k <- k+1
  lambda[2] <- rgamma(1, shape = n+1, scale = 1/(lambda[1]*sdata+3))
  res[k,] <- lambda
  k <- k+1
}
plot(res[,1],res[,2],type="l",xlim=c(0,8),ylim=c(0,8))
```



c) Which type of algorithm is this? Why? (4P)

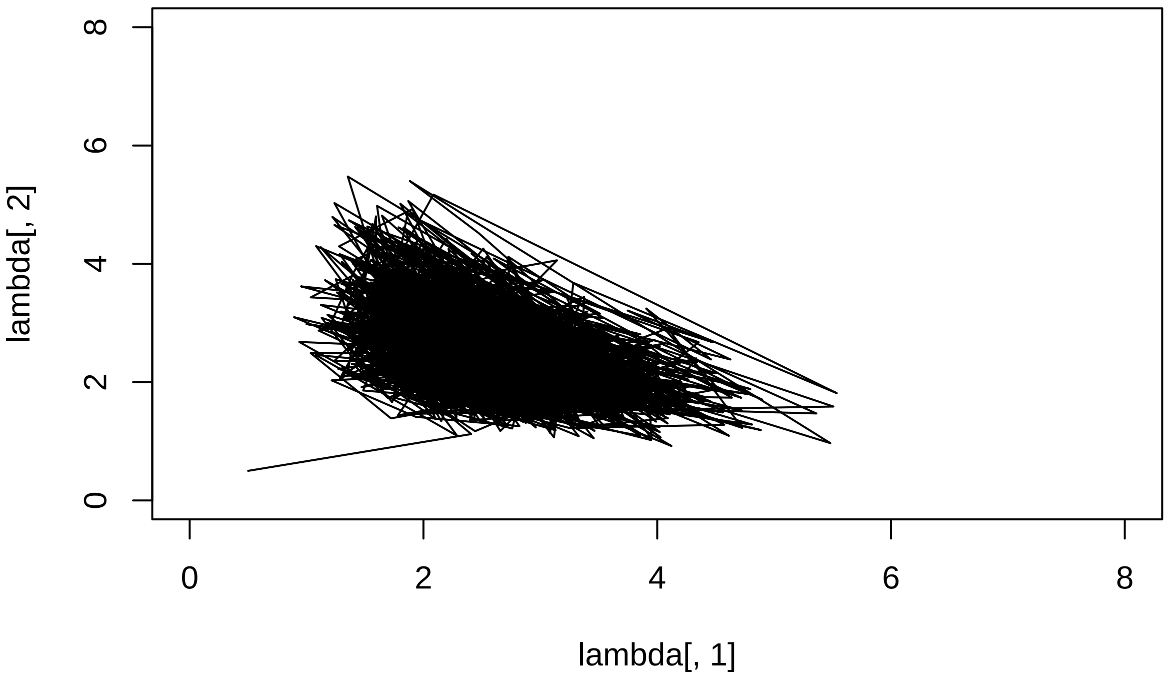
The second algorithm, and a plot of output are given by

```
data <- c(0.08, 0.12, 0.01, 0.01, 0.04, 0.29, 0.12, 0.05, 0.10, 0.01,
          0.14, 0.08, 0.12, 0.44, 0.11, 0.10, 0.19, 0.07, 0.03, 0.06)

sdata <- sum(data)
n <- length(data)

func <- function(lambda){
  res <- 0
  a <- lambda[1]
  b <- lambda[2]
  if(a>0 & b>0)
    res <- (a*b)^n*exp(-a*b*(sdata)-3*a-3*b)
  return(res)
}

nSim=10000
lambda <- matrix(0, nSim, 2)
lambda[1,] <- c(0.5,0.5)
set.seed(1)
tmp <- 0
for(t in 2:nSim){
  lambdaStar <- rnorm(2,mean=lambda[t-1],sd=1.5)
  alpha <- min(1.0,func(lambdaStar)/func(lambda[t-1,]))
  if(runif(1)<alpha && is.finite(alpha)){
    lambda[t,] <- lambdaStar
    tmp <- tmp+1
  } else {
    lambda[t,] <- lambda[t-1,]
  }
}
plot(lambda[,1],lambda[,2],type="l",xlim=c(0,8),ylim=c(0,8))
```



d) Which type of algorithm is this? Why? (4P)

e) Which of the two algorithms above would you prefer? Why? (3P)